Chapter 3: Credit Risk Management

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Credit Migration Approach

With the Bank for International Settlement (BIS) 1998 reform in place, internal models for both general and specific market risk have been implemented at the major G-10 banks and are used every day to report regulatory capital for the trading book.

The current BIS requirements for the "specific risk" component of market risk are subject to broad interpretation.

E.g. "concentration risk", "spread risk", "downgrade risk", and "default risk"

Problems with this piece meal approach

- Spread risk is related to both market risk and credit risk.
- Downgrade risk is a pure credit spread risk.
- The problem of disentangling the market risk- and credit risk- driven components in spread changes
- Default is simply a special case of downgrade
- Changes in market and economic conditions may affect the overall profitability of firms.

Evolution of Credit Migration

- CreditMetrics from JP Morgan in 1997

  The CreditMetrics approach is based on the analysis of credit migration.
- Tom Wilson (1997) proposes an improvement to the credit migration approach by allowing default probabilities to vary with the credit cycle.
- KMV corporation had developed a credit risk methodology and extensive database to assess default probabilities.
- At the end of 1997, Credit Suisse Financial Products (CSFP) release an approach that is based on actuarial science.
- The reduced form approach: the foundation of credit derivative pricing models.

CreditMetrics Framework

CreditMetrics is a methodology that is based on the estimation of the forward distribution of changes in the value of a portfolio of loan and bond-type products at a given time horizon.

The changes in value are related to the migration, upwards and downwards, of the credit quality of the obligor, as well as to default.

- the portfolio distribution is far from being a normal distribution;
- measuring the portfolio effect due to diversification is much more complex than for market risk;
- the information on loans is not as complete as it is for traded instruments such as bonds.

Comparison of Credit Returns and Market Returns

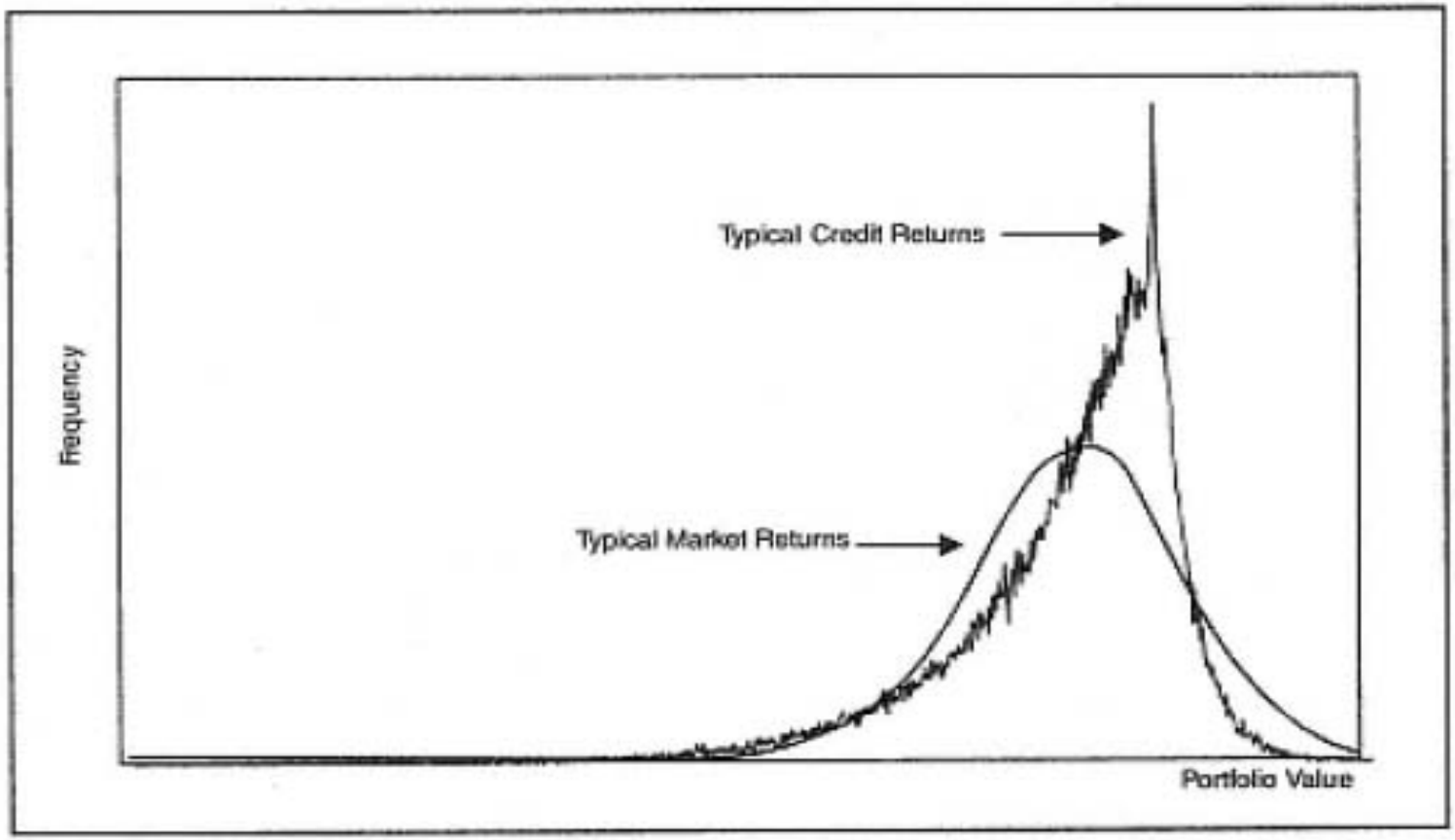

Figure 3.1

Credit VaR for a Bond (Building Block 1)

- Specify a rating system
  The transaction matrix is the key component of the credit VaR model proposed by JP Morgan.
  A strong assumption made by CreditMetrics is that all issuers within the same rating class are homogeneous credit risks. The have the same transition probabilities and the same default probabilities.
- Specify the risk horizon
- Specify the forward discount curve at the risk horizon for each credit category.
- Derive the forward distribution of the changes in portfolio value

Step 1: Specify the transition matrix

The rating categories are chosen from an external or internal rating system.

Assuming that the process for default is Markovian and stationary, the multiplying the one-year transition matrix n times generates the n-year matrix. The n-year default probabilities are simply the values in the column in year n.

However, the realized transition and default probabilities vary quite substantially over the years, depending upon whether the economy is in recession or is expanding.

Transition Matrix

Initial Rating	Rating at Year-End (%)							
	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	90.81	8.33	0.68	0.06	0.12	0	0	0
AA	0.70	90.65	7.79	0.64	0.06	0.14	0.02	0
A	0.09	2.27	91.05	5.52	0.74	0.26	0.01	0.06
BBB	0.02	0.33	5.95	86.93	5.30	1.17	1.12	0.18
BB	0.03	0.14	0.67	7.73	80.53	8.84	1.00	1.06
B	0	0.11	0.24	0.43	6.48	83.46	4.07	5.20
CCC	0.22	0	0.22	1.30	2.38	11.24	64.86	19.79

Figure 3.2

Average Cumulative Default Rates (Percent)

Term	1	2	3	4	5	7	10	15
AAA	0.00	0.00	0.07	0.15	0.24	0.66	1.40	1.40
AA	0.00	0.02	0.12	0.25	0.43	0.89	1.29	1.48
A	0.06	0.16	0.27	0.44	0.67	1.12	2.17	3.00
BBB	0.18	0.44	0.72	1.27	1.78	2.99	4.34	4.70
BB	1.06	3.48	6.12	8.68	10.97	14.46	17.73	19.91
B	5.20	11.00	15.95	19.40	21.88	25.14	29.02	30.65
CCC	19.79	26.92	31.63	35.97	40.15	42.64	45.10	45.10

Figure 3.3

Step 2: Specify the credit risk horizon

The risk horizon is usually set at one year, and is consistent with the transition matrix.

This horizon is arbitrary and is mostly dictated by the availability of the accounting data.

In KMV's framework, any horizon can be chosen. It is assumed that the other characteristics of the borrowers remain constant.

Step 3: Specify the forward pricing model

The valuation of a bond is derived from the zero curve corresponding to the rating of the issuer. All obligors within the same rating class are then marked-to-market using the same curve.

The **spot zero curve** is used to determine the current spot value of the bond.

The **forward price** of the bond one year from the present is derived from the forward zero curve.

Category	Year 1	Year 2	Year 3	Year 4
AAA	3.60	4.17	4.73	5.12
AA	3.65	4.22	4.78	5.17
A	3.72	4.32	4.93	5.32
BBB	4.10	4.67	5.25	5.63
BB	5.55	6.02	6.78	7.27
B	6.05	7.02	8.03	8.52
CCC	15.05	15.02	14.03	13.52

Example

Empirical evidence shows that for high-grade investment bonds, the spreads tend to increase with time to maturity, while for low-grade bonds the spread tends to be wider at the short end.

E.g. If the obligor remains rated BBB, calculate the one year forward price V_{BBB}, of the five year, 6 percent coupon bond.

Year-End Rating	AAA	AA	A	BBB
Value ($)	109.35	109.17	108.64	107.53
Year-End Rating	BB	B	CCC	Default
Value ($)	102.00	98.08	83.62	51.11

Step 4: Derive the forward distribution

When the credit quality changes, the bond value will change as well. The first percentile of the distribution of ΔV is -23.91. It is a much lower value than if we computed the first percentile assuming a normal distribution for ΔV. In that case credit VaR at the 99 percent confidence level would be only -7.43.

Category	Probability of State	Forward Price	Change in Value
AAA	0.02	109.35	1.82
AA	0.33	109.17	1.64
A	5.95	108.64	1.11
BBB	86.93	107.53	0
BB	5.30	102.00	-5.53
B	1.17	98.08	-9.45
CCC	0.12	83.62	-23.91
Default	0.18	51.11	-56.42

Histogram of the One-Year Forward Prices

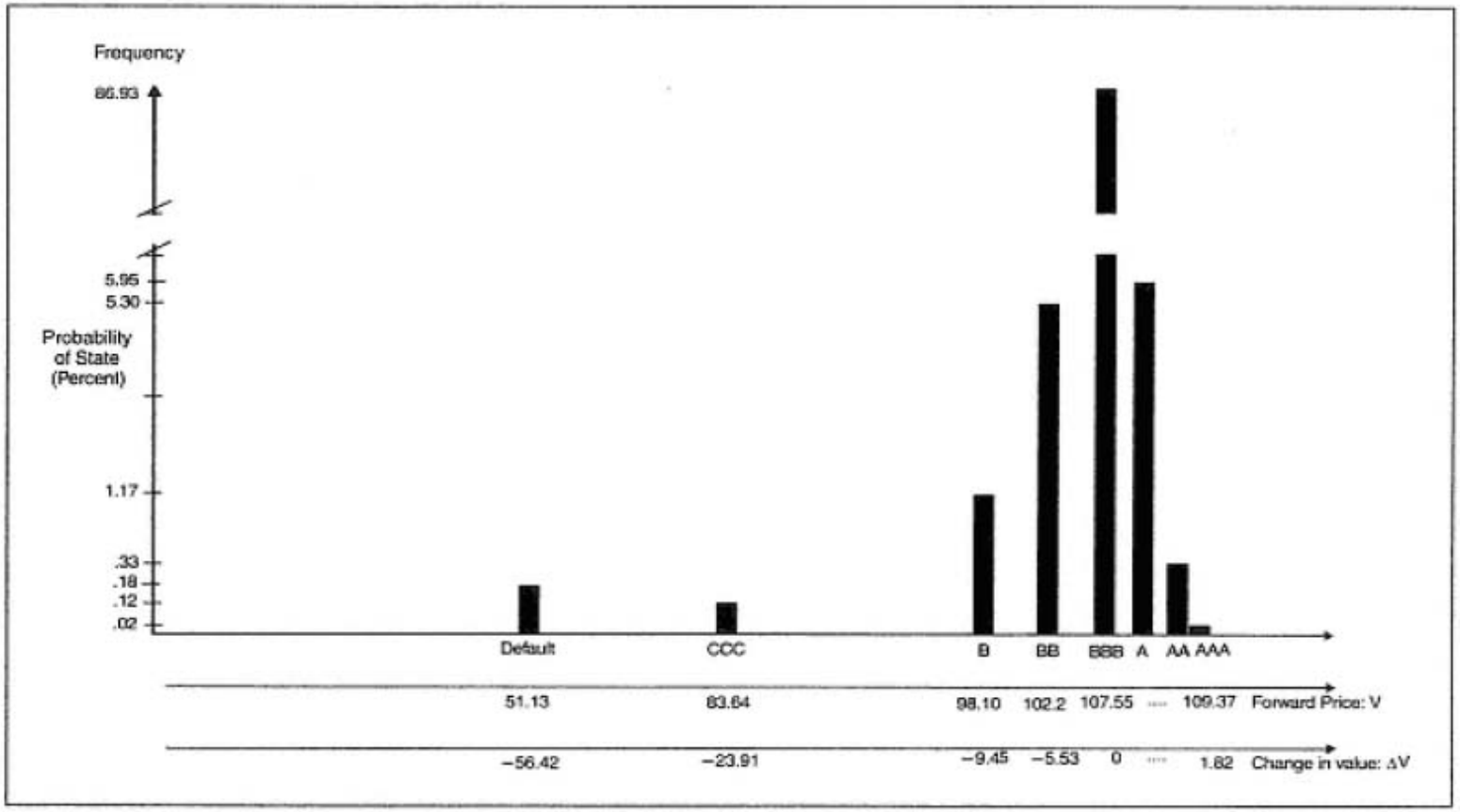

Figure 3.4

Credit VaR for a Loan or Bond Portfolio (Building Block 2)

Assume a portfolio composed of two bonds with a rating of BB and A.

Given the transition matrix, and assuming no correlation between changes in credit quality, we can then derive easily the joint migration probabilities.

Credit VaR for a Loan or Bond Portfolio (Building Block 2)

		Obligor 2							
Obligor		AAA	AA	A	BBB	BB	B	CCC	Default
(BB)		0.09	2.27	91.05	5.52	0.74	0.26	0.01	0.06
AAA	0.03	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00
AA	0.14	0.00	0.00	0.13	0.01	0.00	0.00	0.00	0.00
A	0.67	0.00	0.02	0.61	0.40	0.00	0.00	0.00	0.00
BBB	7.73	0.01	0.18	7.04	0.43	0.06	0.02	0.00	0.00
BB	80.53	0.07	1.83	73.32	4.45	0.60	0.20	0.01	0.05
B	8.84	0.01	0.20	8.05	0.49	0.07	0.02	0.00	0.00
CCC	1.00	0.00	0.02	0.91	0.06	0.01	0.00	0.00	0.00
Default	1.06	0.00	0.02	0.97	0.06	0.01	0.00	0.00	0.00

CreditMetrics Models

Default correlations might be expected to be higher for firms within the same industry or in the same region than for firms in unrelated sectors.

For the simplicity, CreditMetrics makes use of the stock price of a firm as a proxy for its asset value as the true asset value is not directly observable.

CreditMetrics estimates the correlations between the equity returns of various obligor, and then it inters the correlations between changes in credit quality directly from the joint distribution of these equity returns.

Merton's Model

The model assumes a simple capital structure with one type of zero-coupon debt.

It can easily extended to the case where the firm has issued senior and junior debt. In this case, the loss rates for each type of debt are endogenously derived, together with the default probability.

Consider a simple case of a firm with risky assets V, which are financed by equity S and by one debt obligation, maturing at time T with face value of F and market value β. The loan to the firm is subject to credit risk, namely the risk at time T the value of the firm's assets, V_T, will fall below the obligation to the debt holders, F.

Merton's Model

Credit risk exists as long as the probability of default, $Prob(V_T < F)$, is greater than zero. This implies that at time 0, $B_0 < Fe^{-rT}$.

The yield to maturity on the debt, y_T, is higher than the risk-free rate r, where $\pi_T = y_t - r$ denotes the default spread that compensates the bond holders for the default risk that they bear.

If we assume that markets are frictionless, with no taxes, and there is no bankruptcy cost, then the value of the firm's assets is simply the sum of the firm's equity and debt,

$$V_0 \equiv S_0 + B_0$$

Merton's Model

Two assumption:

- the loan is the only debt instrument of the firm;
- the only other source of financing is equity

Time	0	T	
Value of assets	V_0	$V_T \leq F$	$V_T > F$
Bank's position:			
(a) Make a loan	$-B_0$	V_T	F
(b) Buy a put	$-P_0$	$F - V_T$	0
Total	$-B_0 - P_0$	F	F

In equilibrium, it should be that

$$B_0 + P_0 = Fe^{-rT}$$

Merton's Model

The value of the put option is the cost of eliminating the credit risk associated with providing a loan to firm.

If we make the assumptions that are needed to apply the Black-Scholes (1973) model to equity and debt instruments, we have the value of the put as

$$P_0 = -N(-d_1)V_0 + Fe^{-rt}N(-d_2)$$

where P_0 is the current value of the put, $N(.)$ is the cumulative standard normal distribution and

$$d_1 = \frac{\ln(V_0/F) + (r + 1/2\sigma^2)T}{\sigma\sqrt{t}} = \frac{\ln(V_0/Fe^{-rT}) + 1/2\sigma^2 T}{\sigma\sqrt{t}}, d_2 = d_1 - \sigma\sqrt{T}$$

and σ is the standard deviation of the rate of return of the firm's assets.

Merton's Model

Assume that the credit risk, and its costs, is a function of the riskness of the assets of the firm, σ, and this risk is also a function of the time interval until debt is paid back, T.

The cost is also affected by the risk-free interest rate r: the higher r is, the less costly it is to reduce credit risk. The cost is homogeneous function of the leverage ratio, $LR = Fe^{-rT}/V_0$, which means that it stays constant for a scale expansion of Fe^{-rT}/V_0.

$$y_T = -\frac{\ln(B_0/F)}{T} = -\frac{\ln((Fe^{-rT} - P_0)/F)}{T}$$

Given the default spread, $\pi_t = y_T - r$,

$$\pi_T = y_T - r = -\frac{1}{T}\ln\left(N(d_2) + \frac{V_0}{Fe^{-rT}} \cdot N(-d_1)\right)$$

Numerical Example

Default Spread for Corporate Debt				
	Volatility of underlying asset: σ			
LR	0.05	0.10	0.20	0.40
0.5	0	0	0	1.0%
0.6	0	0	0.1%	2.7%
0.7	0	0	0.4%	5.6%
0.8	0	0.1%	1.7%	9.6%
0.9	0.1%	0.9%	4.6%	14.6%
1.0	2.2%	4.6%	9.5%	20.7%

Using BS model, given $V_0 = 100, T = 1, r = 10\%, \sigma = 40\%, LR = 70\%$, we obtain for the value of equity, $S_0 = 33.37$ and the value of the corporate risk debt, $B_0 = 66.64$.

The yield on the loan is equal to $77/66.63 - 1 = 0.156$; i.e., there is a 5.6 percent risk premium on the loan to reflect the credit risk.

Numerical Example

The model also shows that the put value is $P_0 = 3.37$. Hence, the cost of eliminating the credit risk is \$3.37 for \$100 worth of the firm's assets, where the face value of the one-year debt is 77.

This cost drops to 25 cents when volatility decreases to 20 percent, and to 0 for 10 percent volatility. This riskiness of the assets as measure by the volatility σ is a critical factor in determining credit risk.

The yield on the bank's position is:

$$F/(B_0 + P) = 77/(66.63 + 3.37) = 1.10$$

which translates to a riskless yield of 10 percent per annum.

In Merton's model, the firm is assumed to have a very simple capital structure; it is financed by equity S_t, and a single zero-coupon debt instrument maturing at time T, with face value F, and current market value B_t.

	Asset		Liabilities/Equity	
	Risky Asset:	V_t	Debt:	$B_t(F)$
			Equity:	S_t
Total:		V_t		V_t

In this framework, default occurs at the maturity of the debt obligation only when the value of assets is less than the promised payment F to the bondholders.

Distribution of the Firm's Asset Value at maturity

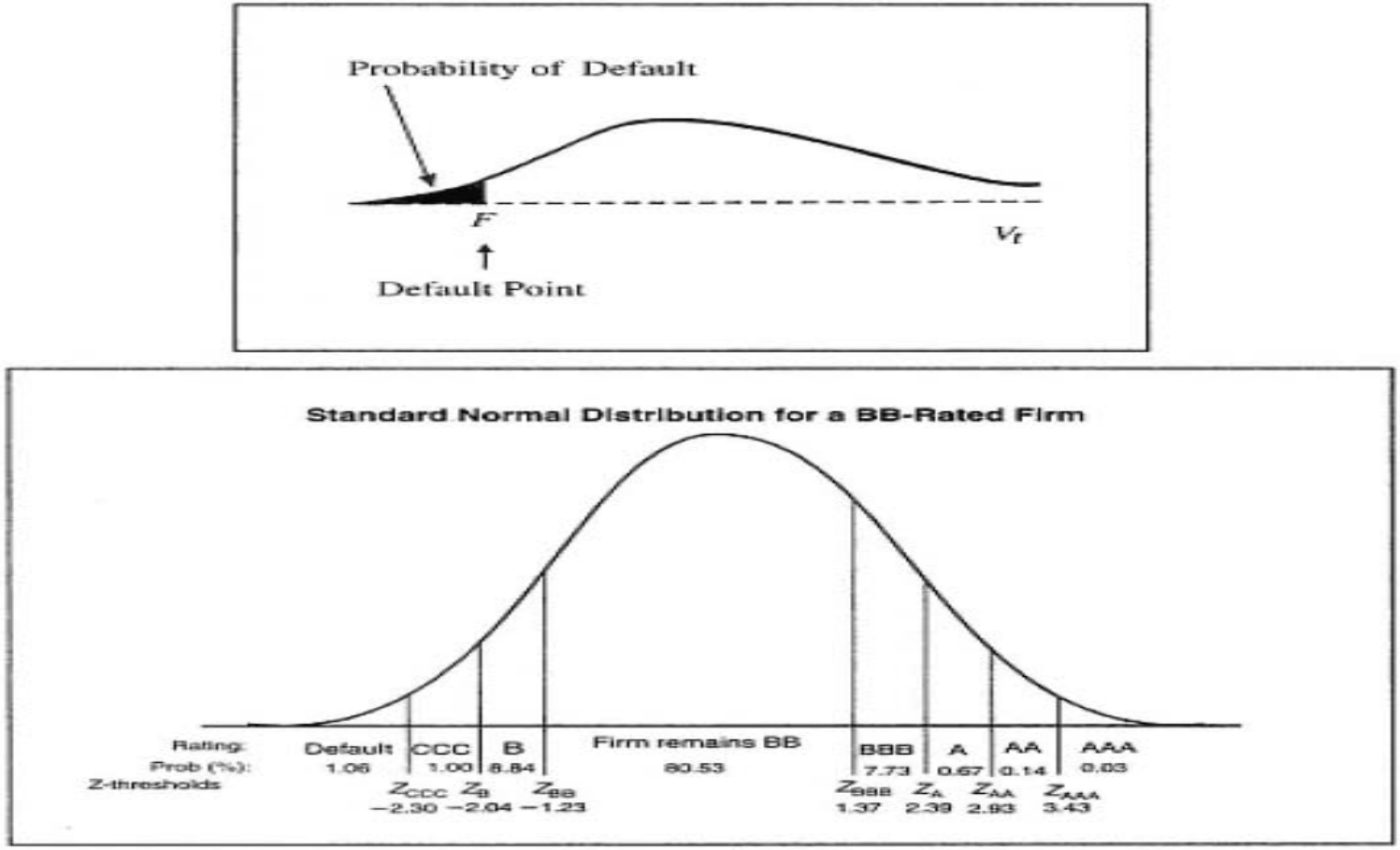

Figure 3.5

CreditMetrics Model

Merton's model is extended by CreditMetrics to include changes in credit quality.

This generalization consists of slicing the distribution of asset returns into bands in such a way that, if we draw randomly from this distribution, we reproduce exactly the migration frequencies as shown in the transition matrices.

This generalization of Merton's model is quite easy to implement. It assumes that the normalized log-returns over any period of time are normally distributed with a mean of 0 and a variance of 1, and that the distribution is the same for all obligors within the same rating category.

Transition Probabilities

Rating in One Year	*A Rated Obligor*		*BB Rated Obligor*	
	Probabilities (%)	Thresholds: Z (σ)	Probabilities (%)	Thresholds: Z (σ)
AAA	0.09	3.12	0.03	3.43
AA	2.27	1.98	0.14	2.93
A	91.05	−1.51	0.67	2.39
BBB	5.52	−2.30	7.73	1.37
BB	0.74	−2.72	80.53	−1.23
B	0.26	−3.19	8.84	−2.04
CCC	0.01	−3.24	1.00	−2.30
Default	0.06		1.06	

Figure 3.6

Default probability and default point

If Prob(Def) denotes the probability of the BB rated obligor defaulting, then the critical asset value V_{Def} is such that:

$$Prbo(Def) = Prob[V_t \leq V_{Def}]$$

which can be translated into a normalized threshold Z_{CCC}, such that the area in the left tail below Z_{CCC} is Prob(Def).

Based on the option pricing model, the critical asset value V_{Def} which triggers default is such that $Z_{CCC} = -d_2$. This critical asset value is also called the "default point".

Note that only the threshold levels are necessary to derive the joint migration probabilities, and these can be calculated without it being necessary to observe the asset value, and to estimate its mean and variance.

CreditMetrics Model

As asset returns are not directly observable, CreditMetrics makes use of equity returns as their proxy.

Yet Using equity returns in this way is equivalent to assuming that all the firm's activities are financed by means of equity. It is a major drawback of the approach, especially when it is being applied to highly leveraged companies. For those companies, equity returns are substantially more volatile, and possible less stationary, than the volatility of the firm's assets.

Assume the correlation is 0.20, the probability that they will remain in the same rating classes is, i.e.

$$Prob(-1.23 < r_{BB} < 1.37, -1.51 < r_A < 1.98) = 0.7365$$

Joint Default Probability

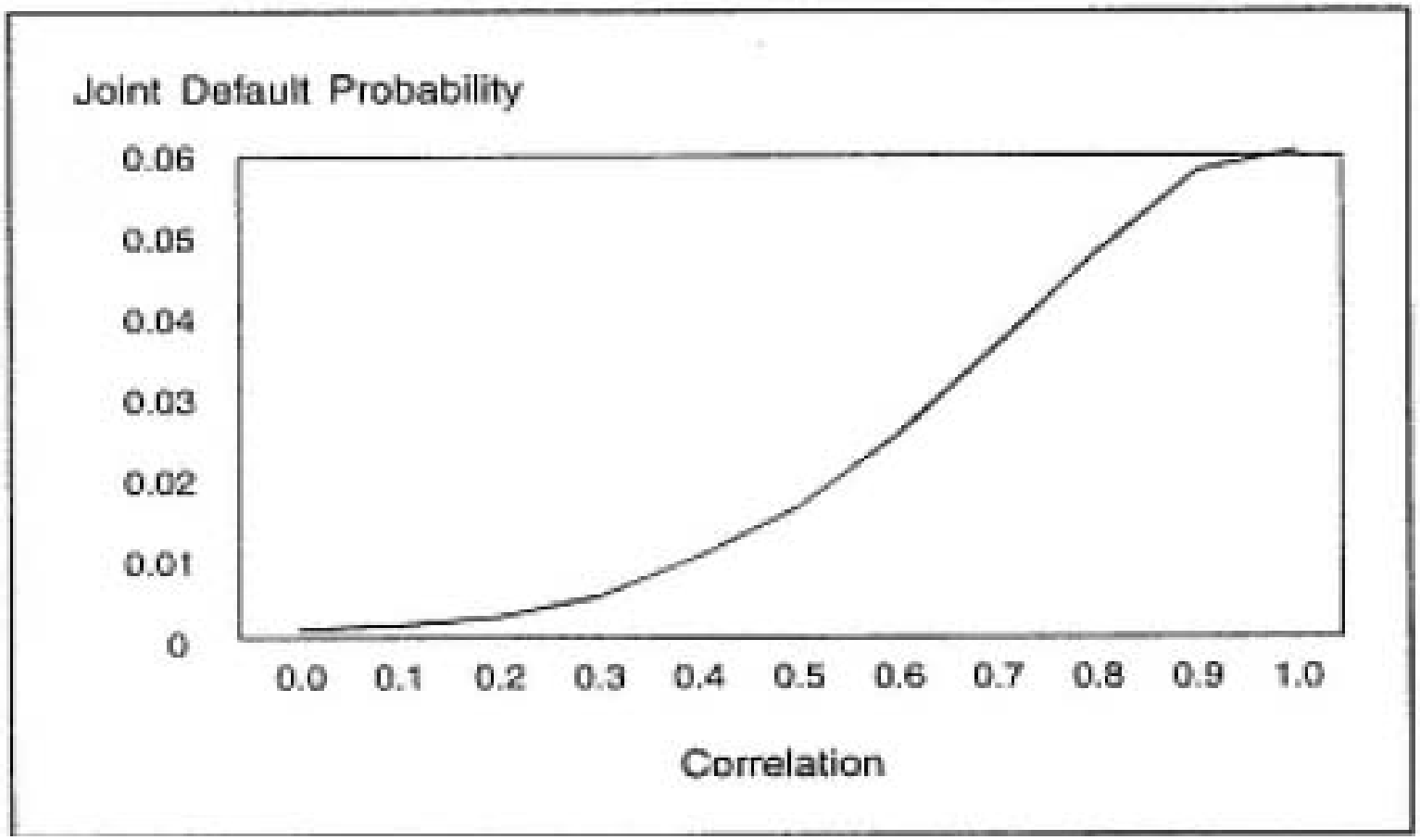

Figure 3.7

Analysis of Credit Diversification

The analytic approach is not practical for large portfolio. Instead, CreditMetrics implements a Monte Carlo simulation to generate the full distribution of the portfolio values at the credit horizon.

- Derivation of the asset return thresholds for each rating category.
- Estimation of the correlation between each pair of obligors' asset returns.
- Generation of asset return scenarios according to their joint normal distribution.
- For each scenario, and for each obligor, the standardized asset return is mapped into the corresponding rating, according to the threshold levels derived in step 1.
- Given the spread curves, which apply for each rating, the portfolio is revalued.
- Repeat the procedure a large number of times, say 100,000 obtain a graph.

Credit VaR and Calculation of the Capital Charge

Economic capital is the financial cushion that a bank uses to absorb unexpected losses.

- $P(c) =$ value of the portfolio in the worst case scenario at the $(I-c)\%$ confidence level
- $FV =$ forward value of the portfolio $= V_0(1+PR)$
- $V_0 =$ current mark-to-market value of the portfolio
- $PR =$ promised return on the portfolio
- $EV =$ expected value of the portfolio $= V_0(1+ER)$
- $ER =$ expected return on the portfolio
- $EL =$ expected loss $= FV - EV$

The capital charge isinstead a function of the unexpected losses

$$Capital = EV - P(c)$$

Credit VaR and Calculation of Economic Capital

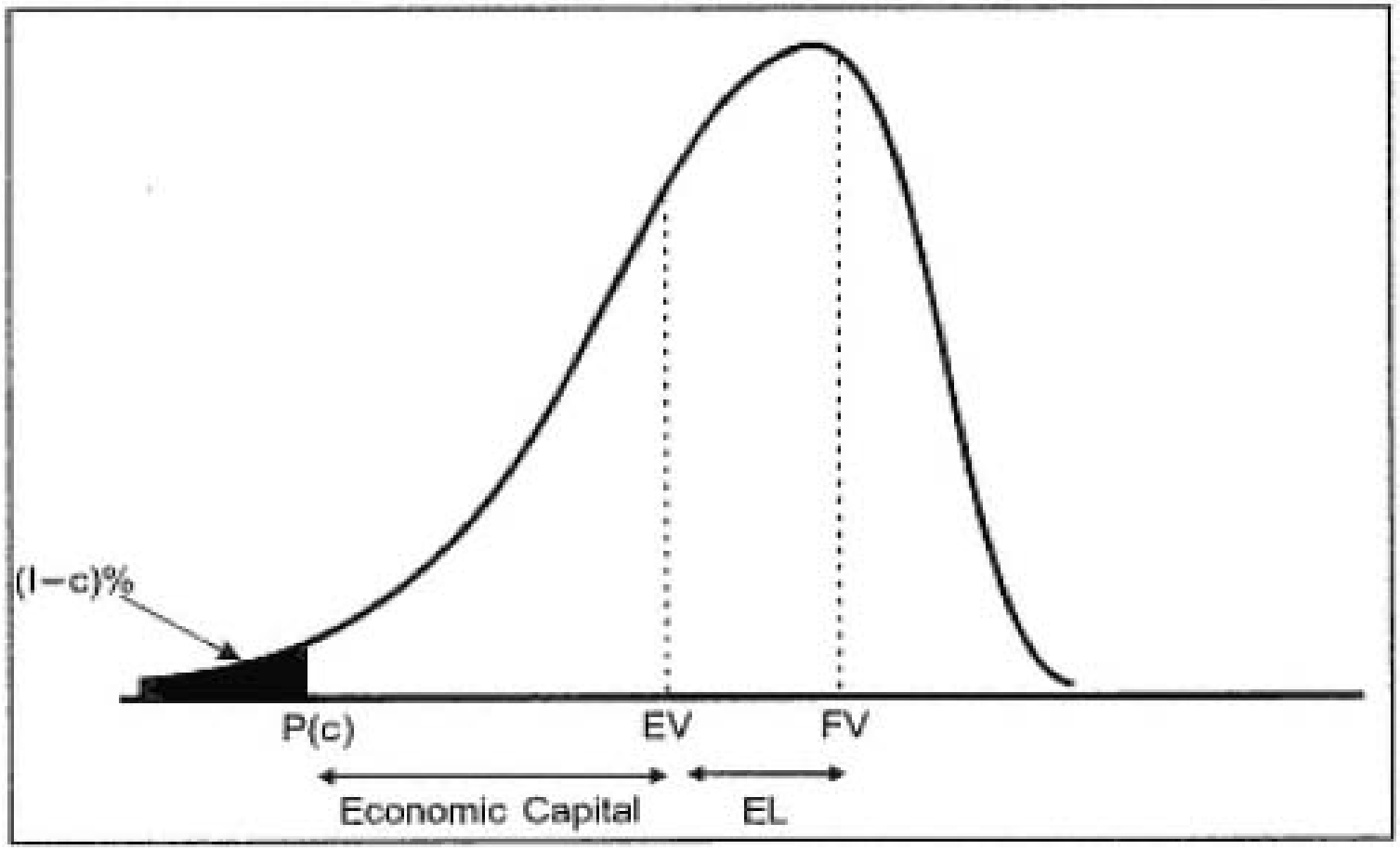

Figure 3.8

CreditMetrics as a Tool: Marginal Risk Measure

CreditMetrics offers the interesting feature of isolating the individual marginal risk contributions to the portfolio.

This is an important proactive risk management tool, as it allows one to identify trading opportunism in the loan/bond portfolio where concentration, and as a consequence overall risk, can be reduced without affecting expected profits.

The same framework can also be used to set up credit risk limits and monitor credit risk exposures in terms of the joint combination of market value and marginal standard deviation.

Risk v.s. Size of Exposures within a Typical Credit Portfolio

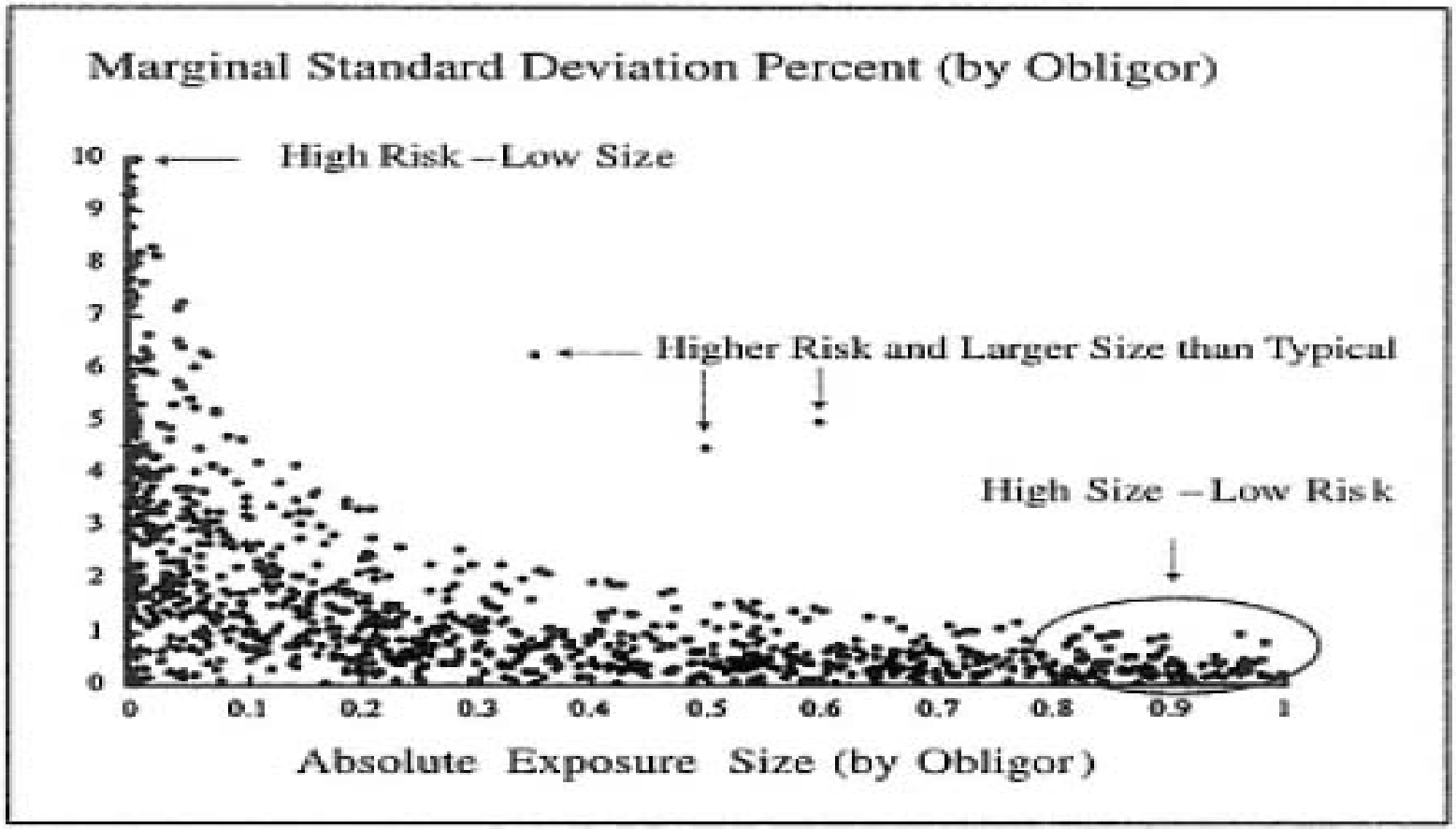

Figure 3.9

Possible Risk Limits for an Example Portfolio

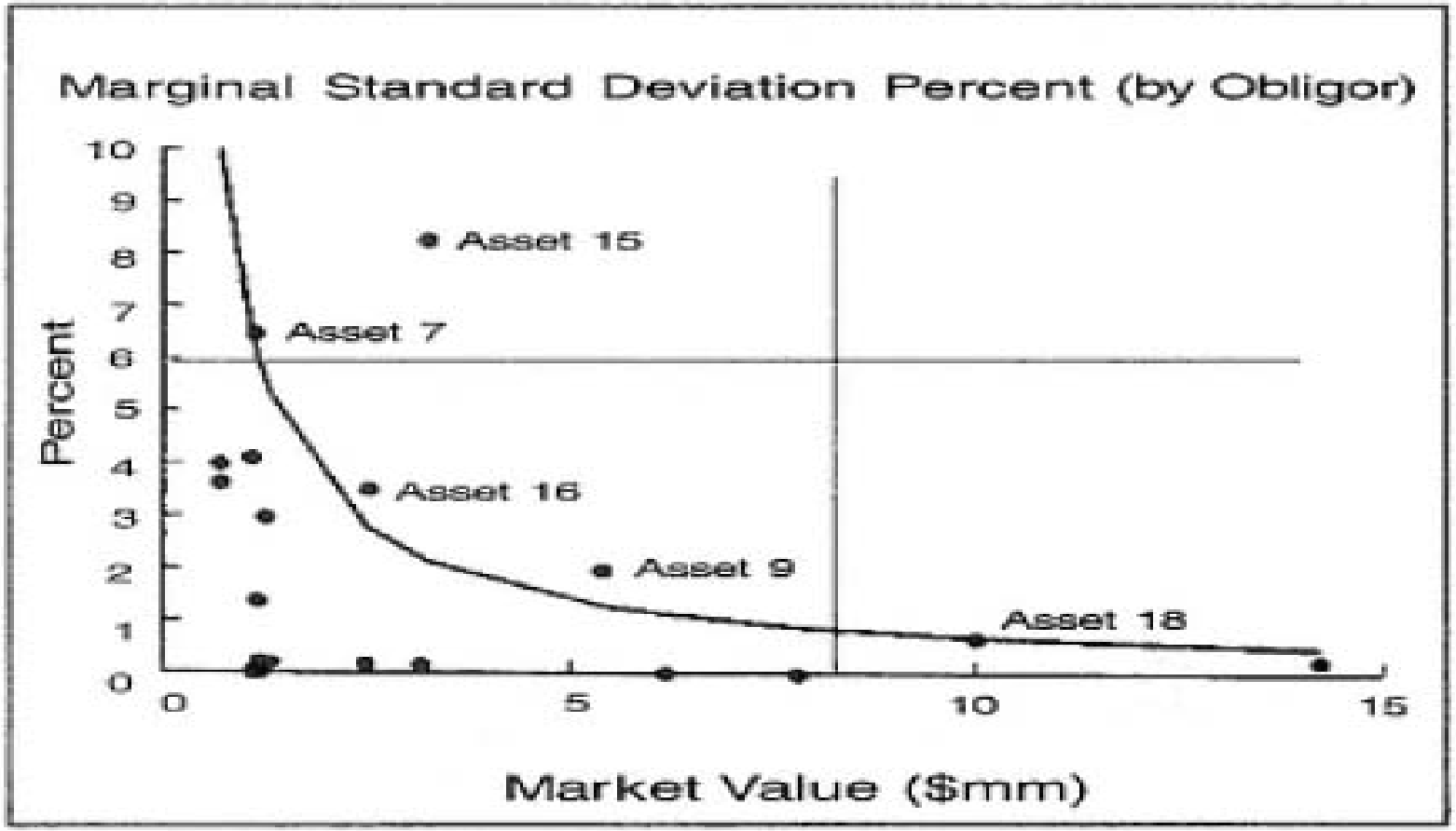

Figure 3.10

Estimation of Asset Correlations (Building Block 3)

Default correlations are derived from asset returns correlations, which in turn are proxied by equity returns correlations.

For a large portfolio of bonds and loans, with thousands of obligors, this still requires the computation of a huge correlation matrix for each pair of obligors.

To reduce the dimensionality of this estimation problem, CreditMetrics uses multifcator analysis. This approach maps each obligor to the countries and industries that are most likely to determine its performance.

Exposures (Building Block 4)

The term "exposures" in CreditMetrics methodology is somewhat misleading since the approach assumes that market risk factors are constant.

So the "exposures" building block is simply the term CreditMetrics gives to the use of the forward pricing model applied to each credit rating.

- Bond-type products
- Derivatives

Example: The swap is a liability to the bank and the swap counterparty is the only party to suffer credit risk.

The Contingent Claim Approach

The CreditMetrics approach is rather appealing as a methodology. Unfortunately it has a major weakness: reliance on ratings transition probabilities that are based on average historical frequencies of defaults and credit migration.

The accuracy of CreditMetrics calculation depends upon two critical assumptions:

- All firms within the same rating class have the same default rate and the same spread curve even when recovery rates differ among obligor;
- The actual default rate is equal to the historical average default rate.

These assumptions cannot be true since we know that default rates evolve continuously, while ratings are adjusted in a discrete fashion.

Introduction to KMV approach

KMV has shown through a simulation exercise that the historical average default rate and transition probabilities can deviate significantly from the actual rates.

KMV has demonstrated that substantial differences in default rates may exist within the same bond rating class, and that the overlap in default probability ranges may be quite large.

E.g. KMV replicated 50,000 times, through a Monte Carlo simulation, Moody's study of default over a 25-year period. For each rating they assumed a fixed number of obligors and assumed that the true probability of default was equal to the reported Moody's average default rate over the 25-year period.

KMV also ran the simulation for several levels of correlation among the asset returns, ranging from 15 percent to 45 percent.

Monte Carlo Simulated Distribution

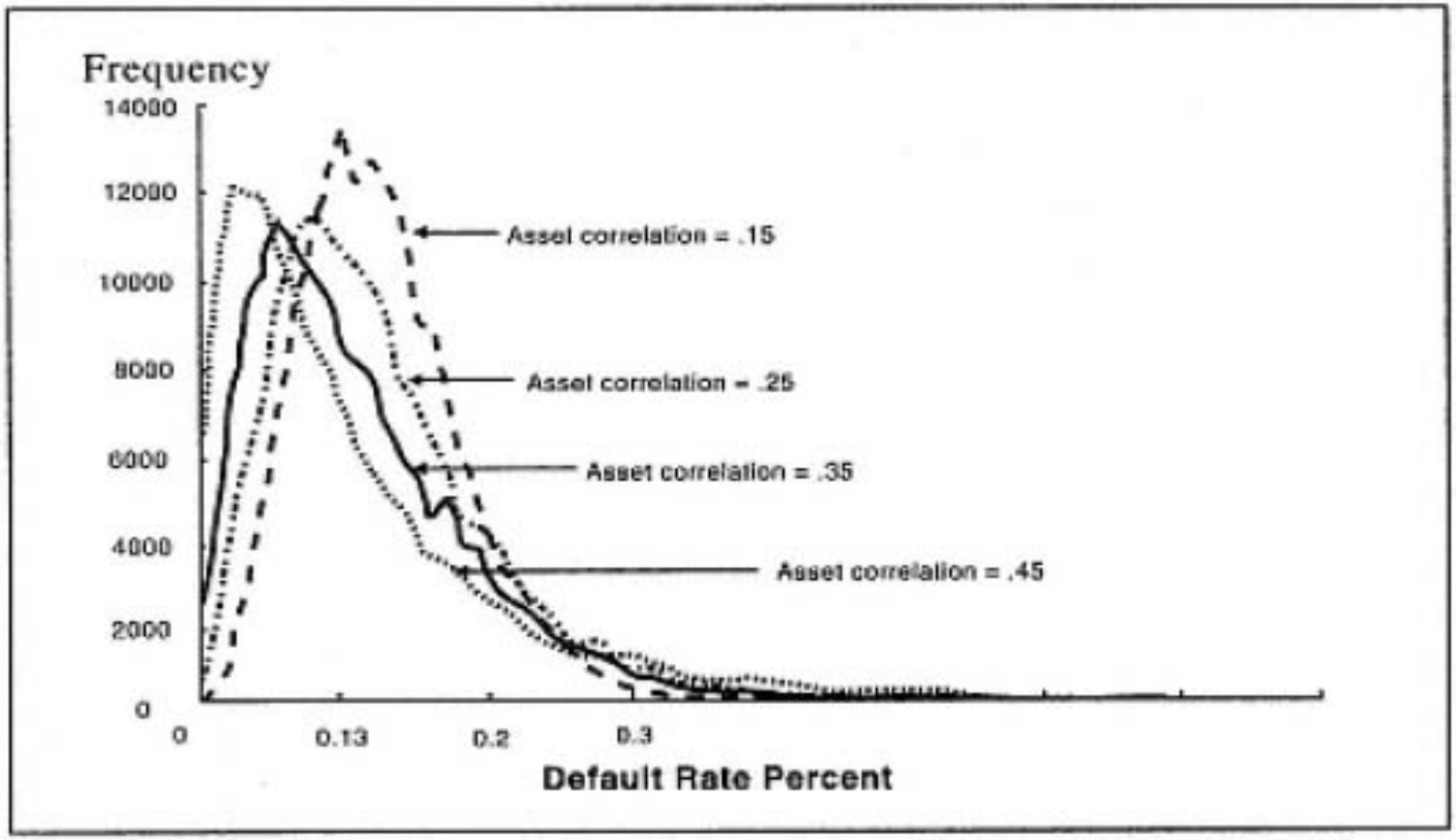

Figure 3.11

Literatures Reviews

- Merton(1974) builds on the limited liability rule which allows shareholders to default on their obligations while they surrender the firm's assets to the various stakeholders, according to prespecified priority rules.
- Longstaff and Schwartz (1995)allows bankruptcy to occur at a random default time. Bankruptcy is triggered the first time the value of the firm's assets falls to some prespecified default boundary; the approach also assumes that the loss in the event of default is exogenous.
- The most recent reduced-form approach, developed independently by Duffie and Singleton (1994), Jarrow and Turnbull (1995), and Jarrow, Lando, and Turnbull (1997). It characterizes bankruptcy as an exogenous process, e.g., as a Markov process in the firm's credit ratings.

Probability of Default and Conditional Expected Recovery Value

Based on Merton's Model and B-S formula, one can extract the probability of default for the loan.

In a risk-neutral world, $N(d_2)$ is the probability that the firm's value at time T will be higher than F, and $1 - N(d_2) = N(-d_2)$ is the probability of default.

By purchasing the put P_0, the bank buys an insurance policy whose premium is the discounted expected value of the expected shortfall in the event of default.

$$P_0 = \left[-\frac{N(-d_1)}{N(-d_2)} V_0 + Fe^{-rT} \right] \cdot N(-d_2)$$

Conditional Expected Recovery Value

The Equation can decompose the premium on the put into three factors.

- The absolute value of the first term inside the bracket is the expected discounted recovery value of the loan, conditional on $V_T \leq F$. It represents the risk-neutral expected payment to the bank in the case where the firm is unable to pay the full obligation F at time T.
- The second term in the bracket is the current value of a riskless bond promising a payment of F at time T.
- The final factor which determines P_0 is the probability of default, $N(-d_2)$.

Numerical Results

Using the numerical example,

$$V_0 = 100, T = 1, r = 10\%, \sigma = 40\%, F = 77, LR = 0.7$$

we can obtain:

$$\begin{aligned}
\text{Discounted expected recovery value} &= \frac{0.135}{0.241} \times 10056.016 \\
\text{Value of riskless bond} &= 77 \times e^{-0.0953} = 70 \\
\text{Expected shortfall} &= 70 - 56.016 = 13.984 \\
\text{Probability of default} &= 24.4\% \\
\text{Cost of default} &= 0.241 \times 13.984 = 3.37
\end{aligned}$$

General Results

For the general case, when the assumption of a risk-neutral world is removed, the probability of default is given by $N(-d_2^1)$ where:

$$d_2^1 = \frac{\ln\left(\frac{V_0}{F}\right) + (\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

and where μ is the expected rate of return on asset V, and V is assumed to be log-normally distributed.

In the previous example, the risk-neutral probability of default is 24.4 percent. If we assume that the discrete time μ is 16 percent, hen the probability of default is 20.5 percent. The expected recovery value is now:

$$\frac{N(-d_1^1)}{N(-d_2^1)} V_0 = \frac{0.110}{0.205} \cdot 100 = 53.7$$

The expected loss EL

The expected loss EL_T in the event of default, at maturity date T:

$$\begin{aligned}
EL_T &= \text{probability of default} \times \text{loss in case of default} \\
&= N(-d_2)F - N(-d_1)V_0 e^{rT} = F - N(d_2)F - N(-d_1)V_0 e^{rT} \\
&= F\left(1 - N(d_2) - N(-d_1)\frac{1}{LR}\right) \\
&= 0.244 \times 77 - 0.137 \times 100 e^{0.0953} = 3.718
\end{aligned}$$

The expected cost of default , expressed in yield, is:

$$\frac{1}{T}\ln\left(\frac{F}{F - EL_T}\right) = -\frac{1}{T}\ln\left(\frac{F\left(N(d_2) + N(-d_1)\frac{V_0}{Fe^{-rT}}\right)}{F}\right) = \pi_T$$

Estimating Credit Risk

A practical problem arises over how easy it is to observe V. In some cases, if both equity and debt are traded, V can be reconstructed by adding the market values of both equity and debt. However, corporate loans are not often traded and so, to all intents and purposes, we can only observe equity.

The question is whether the risk of default can be hedged by trading shares and derivatives on the firm's stock.

The value of equity is

$$S = VN(d_1) - Fe^{-rT}N(d_2)$$

Estimating Credit Risk

A put can be created synthetically by selling short $N(-d_1))$ units of the firm's assets, and buying $Fe^{rT}N(-d_2)$ units of government bonds maturing at T, with face value of F.

If one sells short $N(-d_1)/N(d_1)$ units of the stock S, one effectively creates a short position in the firm's assets of $N(-d_1)$ units, since:

$$-\frac{N(-d_1)}{N(d_1)}S = -VN(-d_1) + Fe^{-rT}N(d_2)\frac{N(-d_1)}{N(d_1)}$$

Therefore, if V is not directly traded or observed, one can create a put option dynamically by selling short the appropriate number of shares. The equivalence between the put and the synthetic put is only valid over short time intervals, and must be readjusted frequently with changes in S and in time left to debt maturity.

Estimating Credit Risk

Using the data from the previous numerical example, $-N(-d_1)/N(d_1) == 0.137/0.863 = -0.159$.

This means that in order to insure against the default of a one-year loan with a maturity value of 77, for a firm with a current market value of assets of 100, the bank should sell short 0.159 of the outstanding equity.

The question now is whether we can use a put option on equity in order to hedge the default risk. It should be remembered that equity itself reflects the default risk, and as a contingent claim its instantaneous volatility σ_s can be expressed as:

$$\sigma_s = \eta_{S,V}\sigma$$

where $\eta_{S,V} = N(d_1)V/S$ is the instantaneous elasticity of equity with respect to the firm's value $\eta_{S,V} = \frac{\partial S}{\partial V}\frac{V}{S}$.

KMV Approach

KMV derives the estimated default frequency or default probability, the EDF, for each obligor based on the Merton (1974) type of model. The probability of default is thus a function of the firm's capital structure, the volatility of the asset returns, and the current asset value.

The EDF is firm specific, and can be mapped onto any rating system to derive the equivalent rating of the obligor. EDFs can be viewed as a "cardinal ranking" of obligors relative to default risk, instead of the more conventional "ordinal ranking" proposed by rating agencies .

KMV's model does not make any explicit reference to the transition probabilities which are already embedded in the EDFs. Indeed, each value of the EDF is associated with a spread curve and an implied credit rating.

An Example

Credit risk in the KMV approach is essentially driven by the dynamics of the asset value of the issuer. Given the capital structure of the firm and once the stochastic process for the asset value has been specified, then the actual probability of default for any time horizon can be derived.

Assume that the firm has a very simple capital structure. It is financed by means of equity S_t and a single zero-coupon debt instrument maturing at time T, with face value F, and current market value B_t.

The firm's balance sheet can be represented as follows: $V_t \equiv B_t(F) + S_t$, where V_t is the value of all the assets. The firm's assets value V_t is assumed to follow a standard geometric Brownian motion.

Assume default only occurs at maturity of the debt, when the value of assets is less than the promised payment F to the bond holders.

Distribution of the Firm's Asset Value

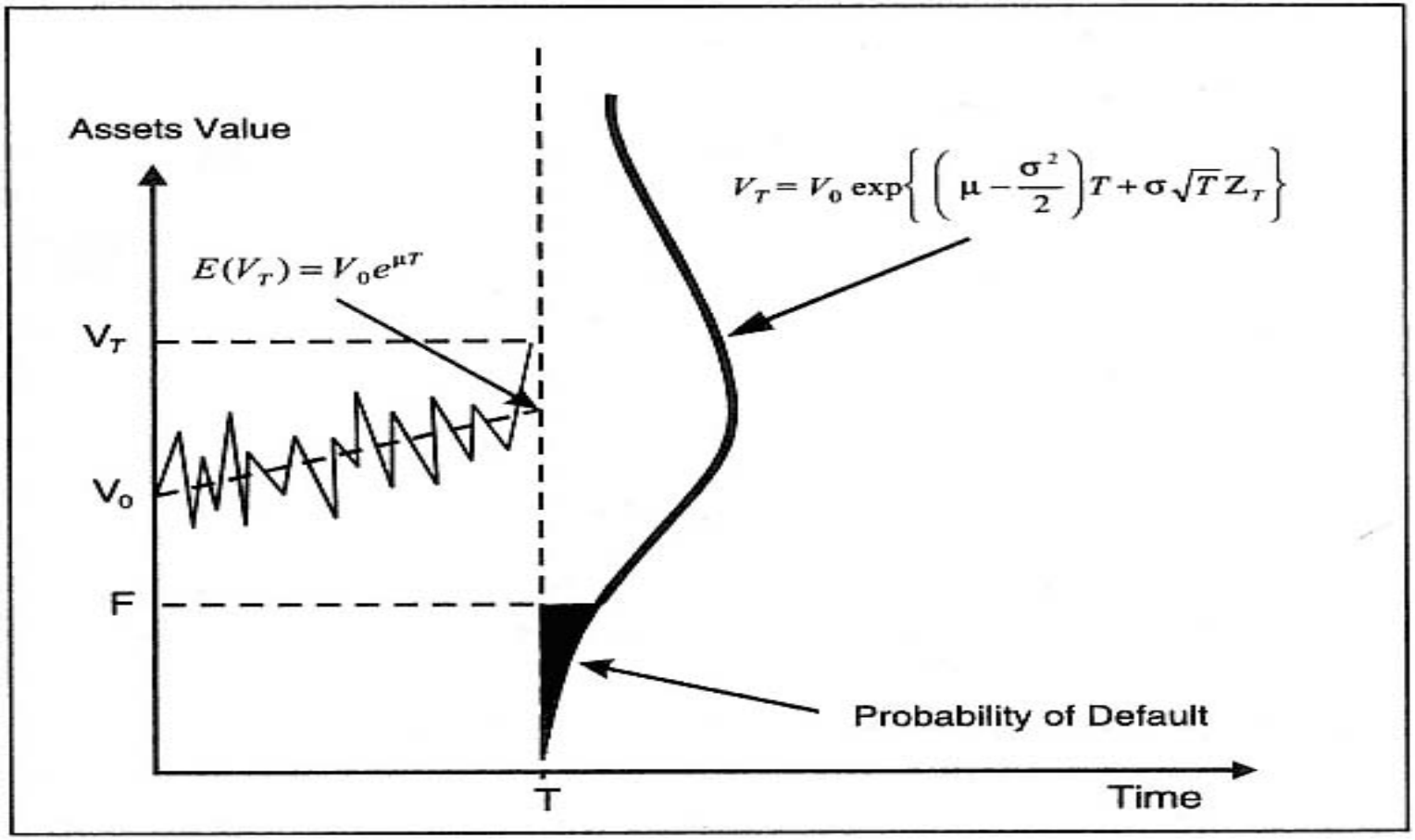

Figure 3.12

KMV Approach

The KMV approach is best applied to publicly traded companies, where the value of the equity is determined by the stock market. The information contained in the firm's stock price and balance sheet can then be translated into an implied risk of default as shown in the next subsections.

The derivation of the actual probabilities of default proceeds in three stages:

- Estimation of the market value and volatility of the firm's assets
- Calculation of the distance to default, which is an index measure of default risk
- Scaling of the distance to default to actual probabilities of default using a default database

Asset Value and the Volatility of Asset Return

In the contingent claim approach to the pricing of corporate securities, the market value of the firm's assets is assumed to be log-normally distributed.

This assumption is quite robust and, according to KMV's own empirical studies, actual data conform quite well to this hypothesis. In addition the distribution of asset returns is stable over time.

In practice, only the price of equity for most public firms is directly observable, and in some cases only part of the debt is actively traded. The alternative approach to asset valuation consists of applying the option pricing model to the valuation of corporate liabilities as suggested in Merton (1974).

Asset Value and Volatility

In order to make their model tractable, KMV assumes that the capital structure of a corporation is composed solely of equity, short-term debt (considered equivalent to cash), long-term debt (in perpetuity), and convertible preferred shares.

Given these simplifying assumptions, it is possible to derive analytical solutions for the value of equity, S, and its volatility, σ_S:

$$\begin{aligned} S &= f(V, \sigma, LR, c, r) \\ \sigma_S &= g(V, \sigma, LR, c, r) \end{aligned}$$

where LR denotes the leverage ratio in the capital structure, c is the average coupon paid on the long-term debt, and r is the risk-free interest rate.

Calculation of the Distance to Default

In the option pricing framework default, or equivalently bankruptcy, occurs when the asset value falls below the value of the firm's liabilities.

Bankruptcy describes the situation in which the firm is liquidated, and the proceeds from the asset sale are distributed to the various claim holders according to prespecified priority rules.

Default, on the other hand, is usually defined as the event when a firm misses a payment on a coupon and/or the reimbursement of principal at debt maturity.

Calculation of the Distance to Default

Using a sample of several hundred companies, KMV observed that firms default when the asset value reaches a level that is somewhere between the value of total liabilities and the value of short-term debt.

For all these reasons, KMV implements an intermediate phase before computing the probabilities of default.

KMV computes an index called ”distance to default” (DD). DD is the number of standard deviations between the mean of the distribution of the asset value, and a critical threshold, the ”default point,” set at the par value of current liabilities including short-term debt to be serviced over the time horizon, plus half the longterm debt.

Distance to Default (DD)

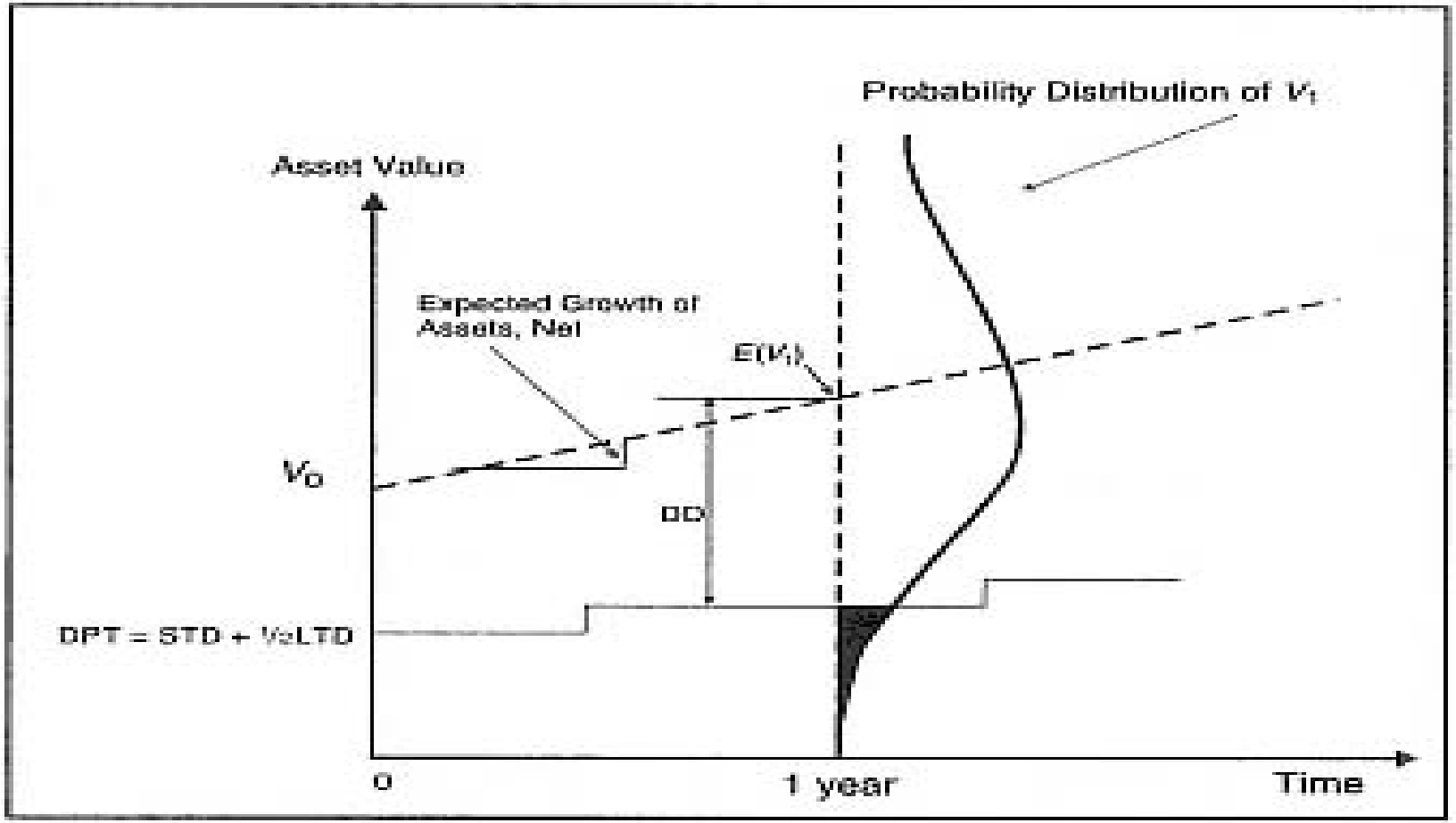

Figure 3.13

Calculation of the Distance to Default

Given the log-normality assumption of asset values, the distance to default, expressed in units of asset return standard deviation at time horizon T, is:

$$DD = \frac{\ln \frac{V_0}{DPT_T} + (\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

where

- V_0 = current market value of assets
- DPT_T = default point at time horizon T
- μ = expected return on assets, net of cash outflows
- σ = annualized asset volatility

Probabilities of Default from the DD

This last phase consists of mapping the "distance-to-default" (DD) to the actual probabilities of default, for a given time horizon. KMV calls these probabilities "expected default frequencies," or EDFs.

Example 1:

Current market value of assets:	$V_0 = 1000$
Net expected growth of assets per annum:	20%
Expected asset value in one year:	$V_0(1.20) = 1200$
Annualized asset volatility, σ:	100
Default point:	800

Then distance-to-default is $\frac{1200-800}{100} = 4$.

Assume that among the population of all the firms with a DD of 4 at one point in time, e.g., 5000, 20 defaulted one year later; then EDF=20/5000=0.4%.

The implied rating for this probability of default is BB^+.

Mapping of "Distance to Default" into the EDFs

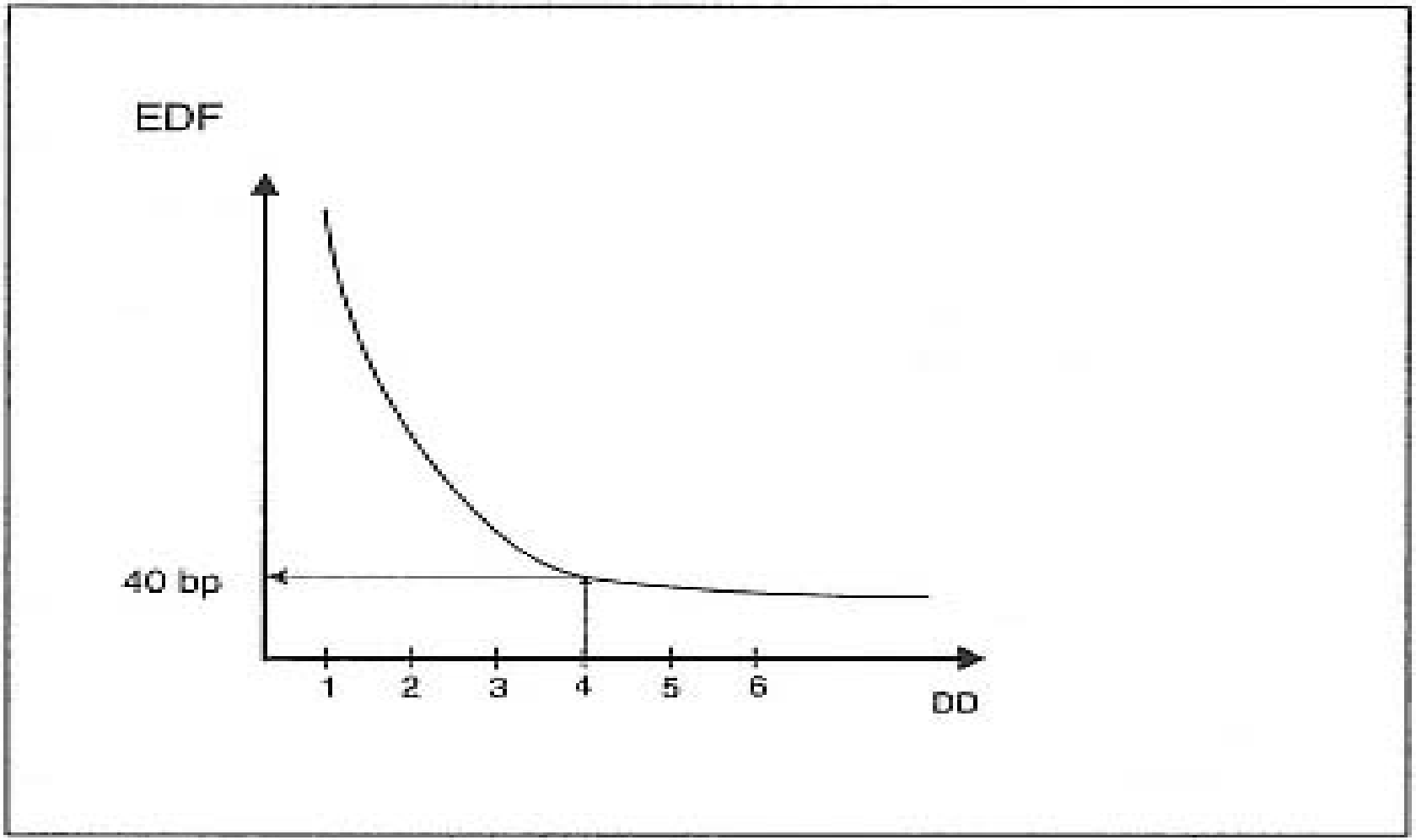

Figure 3.14

EDF as a Predictor of Default

KMV has provided a ”Credit Monitor” service for estimated EDFs since 1993. EDFs have proved to be a useful leading indicator of default, or at least of the degradation of the creditworthiness of issuers. When the financial situation of a company starts to deteriorate, EDFs tend to shoot up quickly until default occurs.

KMV has analyzed more than 2000 U.S. companies that have defaulted or entered into bankruptcy over the last 20 years. These firms belonged to a large sample of more than 100,000 company-years with data, provided by Compustat. In all cases KMV has shown a sharp increase in the slope of the EDF a year or two prior to default.

Changes in EDFs tend to anticipate—by at least one year—the downgrading of the issuer by rating agencies such as Moody’s and Standard & Poor’s.

EDFs and Ratings

Standard & Poor's risk ratings represent default probabilities only, while Moody's factors also include a measure of the probability of loss.

Within any rating class the default probabilities of issuers are clustered around the median. However, the average default rate for each class is considerably higher than the default rate of the typical firm. This is because each rating class contains a group of firms which have much higher probabilities of default, due to the exponential change in default rates as default risk increases.

EDFs and Risk Rating Comparisons

EDF	S&P	Moody's Factors	CIBC	Nationsbank	SBC
2 to 4 bp	>=AA	>=Aa2	1	AAA	C1
4 to 10 bp	AA/A	A1	2	AA	C2
10 to 19 bp	A/BBB+	Baa1	3	A	C3
19 to 40 bp	BBB+/BBB-	Baa3	4	A/BB	C4
40 to 72 bp	BBB-/BB	Ba1	5	BBB/BB	C5
72 to 101 bp	BB/BB-	Ba3	6	BB	C6
101 to 143 bp	BB-/B+	B1	7	BB	C7
143 to 202 bp	B+/B	B2	8	BB/B	C8
202 to 345 bp	B/B-	B2	9	B	C9

Figure 3.15

EDFs and Ratings

Three consequences follow from this analysis.

- First, since the rating agencies are slow to change their ratings, the historical frequency of remaining in a rating class should overstate the true probability of maintaining a particular credit quality.
- Second, the average historical probability of default overstates the true probability of default for typical firms within each rating class, due to the difference between the mean and the median default rates.
- Third, if both the probability of staying in a given rating class and the probability of default are too large, then the transition probabilities must be too small.

KMV Results

Transition Matrix Based on Actual Rating Changes

Initial Rating	Rating at Year-End (%)							
	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	90.81	8.33	0.68	0.06	0.12	0	0	0
AA	0.70	90.65	7.79	0.64	0.06	0.14	0.02	0
A	0.09	2.27	91.05	5.52	0.74	0.26	0.01	0.06
BBB	0.02	0.33	5.95	86.93	5.30	1.17	1.12	0.18
BB	0.03	0.14	0.67	7.73	80.53	8.84	1.00	1.06
B	0	0.11	0.24	0.43	6.48	83.46	4.07	5.20
CCC	0.22	0	0.22	1.30	2.38	11.24	64.86	19.79

Transition Matrix Based on Actual Rating Changes

Initial Rating	Rating at Year-End (%)							
	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	90.81	8.33	0.68	0.06	0.12	0	0	0
AA	0.70	90.65	7.79	0.64	0.06	0.14	0.02	0
A	0.09	2.27	91.05	5.52	0.74	0.26	0.01	0.06
BBB	0.02	0.33	5.95	86.93	5.30	1.17	1.12	0.18
BB	0.03	0.14	0.67	7.73	80.53	8.84	1.00	1.06
B	0	0.11	0.24	0.43	6.48	83.46	4.07	5.20
CCC	0.22	0	0.22	1.30	2.38	11.24	64.86	19.79

Figure 3.16

Asset Return Correlation Model

CreditMetrics and KMV derive asset return correlations from a structural model that links correlations to fundamental factors.

By imposing a structure on the return correlations, sampling errors inherent in simple historical correlations are avoided, and a better accuracy in forecasting correlations is achieved.

In addition, there is a practical need to reduce dramatically the number of correlations that need to be calculated. Assume that a bank is dealing with $N = 100$ different counterparties. Then, we have $N(N-1)/2 = 4950$ different correlations to estimate. This is a staggering number. Multifactor models of asset returns reduce the number of correlations that have to be calculated to the limited number of correlations between the common factors that affect asset returns.

Asset Return Generating Process

It is assumed that the firm's asset returns are generated by a set of common, or systematic, risk factors and idiosyncratic factors. The idiosyncratic factors are either firm specific, or country or industry specific, and do not contribute to asset return correlations (since they are not correlated with each other, and not correlated with the common factors).

Asset return correlations between two firms can be explained only in terms of the factors common to all firms. Only the risks associated with the idiosyncratic risk factors can be diversified away through portfolio diversification, while the risk contribution of the common factors is, on the contrary, nondiversifiable.

Asset Return Generating Process

Assume the asset return generating process for all firms is:

$$r_k = \alpha_k + \beta_{1k} I_1 + \beta_{2k} I_2 + \varepsilon_k, \text{for } k = 1, \cdots, N$$

where

N	=	number of obligors (firms)
r_k	=	asset return for firm k (r_{v_k} denoted r_k)
α_k	=	component of asset return independent of common fac
I_1, I_2	=	common factors
β_{1k}, β_{2k}	=	expected changes in r_k, given a change in I_1 and I_2
ε_k	=	idiosyncratic risk factor with zero mean,
		and assumed to be uncorrelated all the common facto

Results in Portfolio Theory

Using elementary statistical calculus, we can derive results that are well known in portfolio theory:

$$\begin{aligned}
\sigma_k^2 &= \beta_{1k}^2 var(I_1) + \beta_{2k}^2 var(I_2) + var(\varepsilon_k^2) + 2\beta_{1k}\beta_{2k} cov(I_1, I_2) \\
\sigma_{ij} &= \beta_{1i}\beta_{1j} 2var(I_1) + \beta_{2i}\beta_{2j} 2var(I_2) + (\beta_{1i}\beta_{2j} + \beta_{2i}\beta_{1j}) cov(I_1, I_2)
\end{aligned}$$

If we denote by ρ_{ij} the asset return correlation between firm i and firm j, then:

$$\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$$

To derive the asset return correlation between any number of firms, we only need to estimate the β_{ik}, and the covariance matrix for the common factors.

For K common factors, the number of parameters to be estimated is $KN + K(K-1)/2$.

Structure of the Factors

CreditMetrics and KMV propose relatively similar models, so here we will present only the KMV model.

KMV constructs a three-layer factor structure model.

- First level: a composite company-specific factor, which is constructed individually for each firm based on the firm's exposure to each country and industry
- Second level: country and industry factors
- Third level: global, regional, and industrial sector factors

Factor Model for Asset Return Correlations

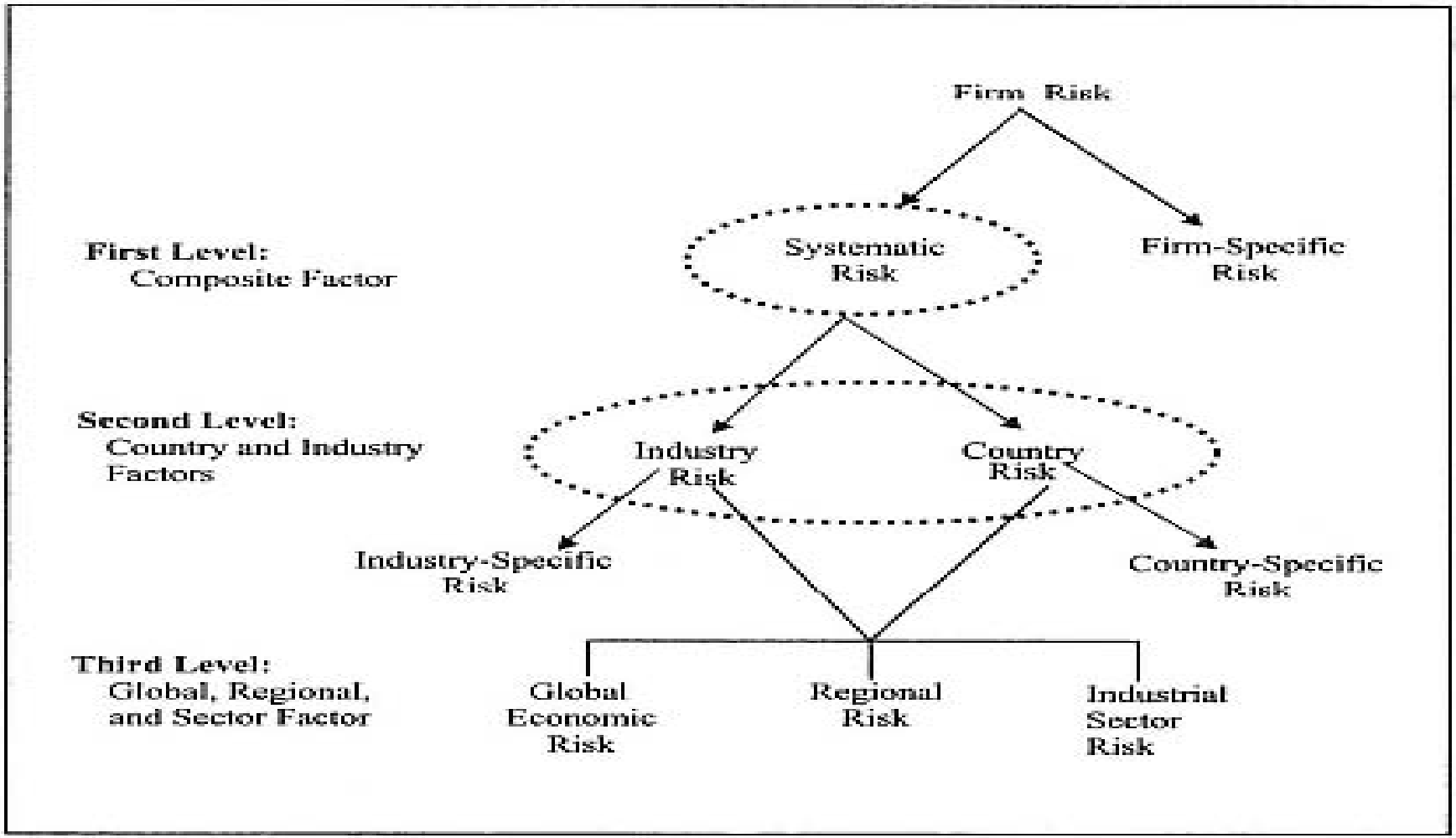

Figure 3.17

A Three-Layer Factor Structure Model

The first level of the structure divides between firm-specific, or idiosyncratic, risk and common, or systematic, risk. Systematic risk is captured by a single composite index and industry factors defined at the second level of the structure:

$$r_k = \beta_k CF_k + \varepsilon_k, \text{for } k = 1, \cdots, N$$

where

r_k	=	asset return for firm k
CF_k	=	composite factor for firm k
β_k	=	firm k's response to the composite factor,
ε_k	=	firm k's specific risk factor

The Composite Factor

The composite factor is constructed as the sum of the weighted country and industry factors specified at the second level of the structure:

$$CF_k = \sum_m \alpha_{km} C_m + \sum_n \alpha_{kn} I_n$$

where

$$\begin{aligned}
C_m &= \text{rate of return on country risk factor } m \\
I_n &= \text{rate of return on industry risk factor } n \\
\alpha_{km} &= \text{weight of firm } k \text{ in country } m\text{, with constraint } \textstyle\sum_m \alpha_{km} = 1 \\
\alpha_{kn} &= \text{weight of firm } k \text{ in industry } n\text{, with constraint } \textstyle\sum_n \alpha_{kn} = 1
\end{aligned}$$

Example

Consider a Canadian firm that has two lines of business, and assume that the data are extracted from Compustat:

Business Line	SIC	Assets	Sales
Lumber and forestry	2431	35%	45%
Paper production	2611	65%	55%
Total		100%	100%

The weight for lumber and forestry is: $40\% = (35\% + 45\%)/2$ and for paper it is: $60\% = (65\% + 55\%)/2$.

The composite factor can be written as:

$$CF = 1.0C_{\text{Canada}} + 0.6I_{\text{paper}} + 0.4I_{\text{lumber}}$$

The Third Level of the Factor Structure

The risk of countries and industries is further decomposed into systematic and idiosyncratic components. The systematic component is captured by basic factors such as global economic effect, regional factor effect, and sector factor effect.

While the common factor is firm specific, the third-level factors are the same for all countries and all industries:

Country Return = Global economic effect + Regional factor effect + Sector factor effect + Country-specific risk

Industry Return = Global economic effect + Regional factor effect + Sector factor effect + Industry-specific risk